

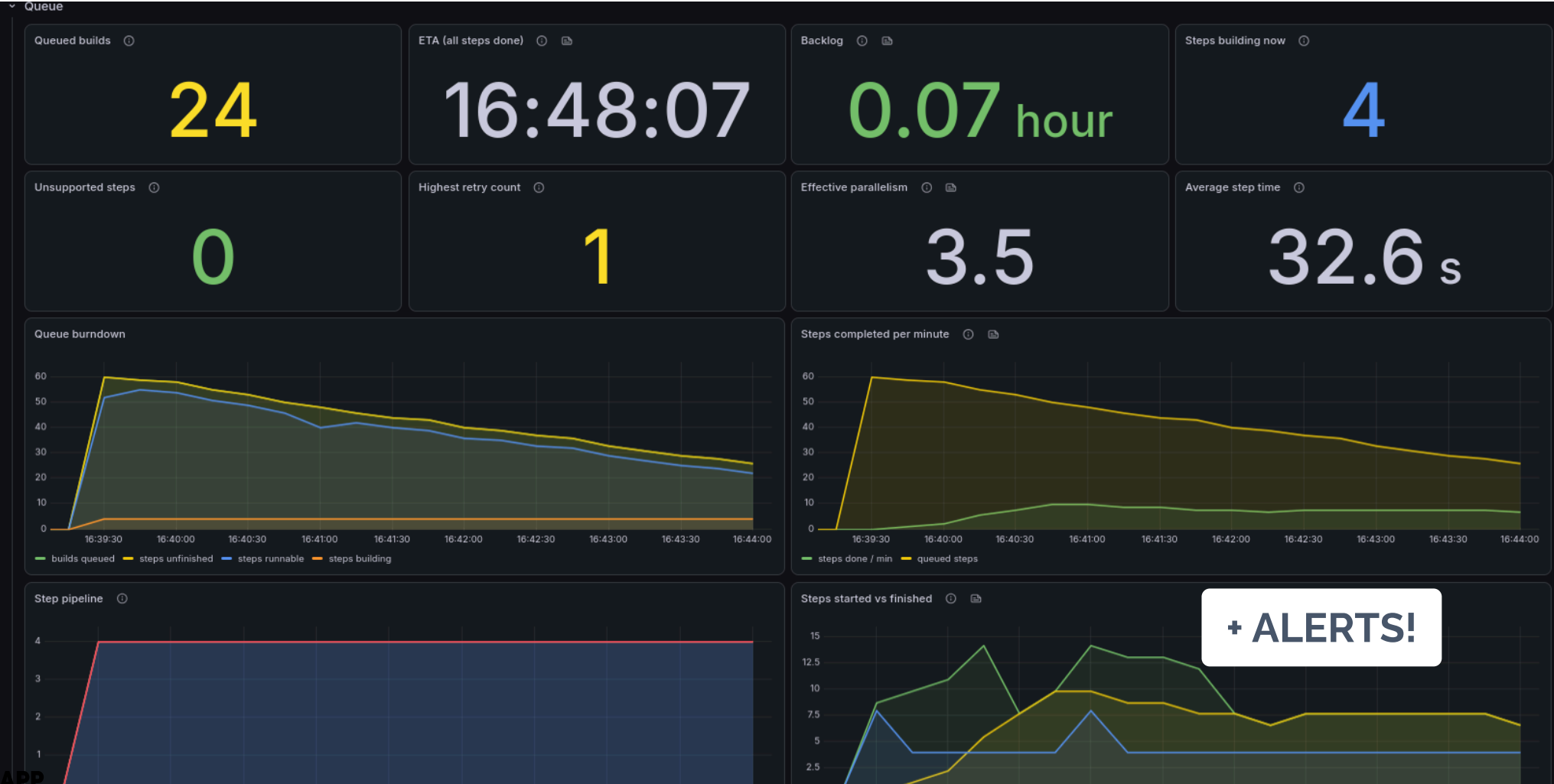
HOW LONG UNTIL IT'S DONE? YOUR OWN HYDRA DASHBOARD

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Are ya winning, son Hydra?



Dramatis personae



Figure 1: *Prometheus*
(Heinrich Fügen, 1817)



Figure 2: *Jason*
(John William Waterhouse, 1907)



Figure 3: *Hydra*
(Gustave Moreau, 1876)

“Hydra” metrics

Me Mom, can we have **Hydra metrics**?

Mom No, we have **Hydra metrics** at home.

Hydra metrics at home —

```
http_request_duration_seconds_bucket  
http_request_duration_seconds_count  
http_request_duration_seconds_sum  
http_request_size_bytes_bucket  
http_request_size_bytes_count  
http_request_size_bytes_sum  
http_requests_total  
http_response_size_bytes_bucket  
http_response_size_bytes_count  
http_response_size_bytes_sum
```

Hydra “metrics”

```
{ "status": "up", "time": 1754630000, "uptime": 3600, "nrQueuedBuilds": 42,
  "nrUnfinishedSteps": 55, "nrStepsBuilding": 8, "nrStepsWaiting": 0,
  "nrUnsupportedSteps": 0, "bytesSent": 123456789, "bytesReceived": 987654321,
  "nrBuildsRead": 100, "buildReadTimeMs": 5000, "buildReadTimeAvgMs": 50,
  "nrBuildsDone": 58, "nrStepsStarted": 70, "nrStepsDone": 62, "nrRetries": 3,
  "maxNrRetries": 2, "nrQueueWakeups": 12, "nrDispatcherWakeups": 34,
  "dispatchTimeMs": 456, "dispatchTimeAvgMs": 13.4, "nrDbConnections": 4,
  "nrActiveDbUpdates": 1, "totalStepTime": 12345, "totalStepBuildTime": 11111,
  "avgStepTime": 199.1, "avgStepBuildTime": 179.2, "machines": { "ssh://root@
builder1": { "enabled": true, "systemTypes": ["x86_64-linux"],
  "supportedFeatures": [], "mandatoryFeatures": [], "nrStepsDone": 40,
  "currentJobs": 4, "disabledUntil": 0, "lastFailure": 0,
  "consecutiveFailures": 0, "totalStepTime": 8000, "totalStepBuildTime": 7000,
  "avgStepTime": 200.0, "avgStepBuildTime": 175.0 }, ... }
```

Two workarounds

NixOS infra: ~600 line custom `hydra_queue_python_reexporter.py`

```
yield self.trivial_gauge(  
    "steps_queued",  
    "Current number of steps for the build queue",  
    self.destructive_read("nrUnfinishedSteps"),  
)  
yield self.trivial_counter(  
    "steps_done",  
    "Total count of steps completed",  
    self.destructive_read("nrStepsDone"),  
)
```

Alternative: NixOS module `services.prometheus.exporters.json`

```
(gauge    "hydraqueue_runner_steps_unfinished" "{ .nrUnfinishedSteps }")  
(counter  "hydraqueue_runner_steps_finished"   "{ .nrStepsDone }")
```

Some ideas for metrics and alerts

👁👁 **queue depth** `hydraqueueerunner_builds_unfinished`

👁👁 **currently building** `hydraqueueerunner_steps_building`

👁👁 **ETA** `(time() + (hydraqueueerunner_steps_unfinished / clamp_min(rate(hydraqueueerunner_steps_finished[1h]), 0.1))) * 1000`

🔥 **HydraBuildersDisabled** `hydra_machine_disabled_until > time()`

🔥 **HydraNoBuildersLeft** `sum(hydra_machine_enabled) == 0`

DEMO

Try it at home

```
$ nix run github:applicative-systems/hydra-dashboard#demo
```

or

```
{  
  imports = [ hydra-dashboard.nixosModules.default ];  
  
  services.hydra-metrics-classic = {  
    enable = true;  
    hydraStatusUrl = "https://hydra.example.com/queue-runner-status";  
    machines = [ "ssh://root@builder1" "ssh://root@builder2" ];  
    systemTypes = [ "x86_64-linux" ];  
    jobsets = [ "myproject:main" ];  
  };  
}
```